

Betting big on solar power

With its growing electricity demand, India has turned its focus on solar energy based power generation. The country is already amongst the world's most active players in renewable energy utilisation, especially wind energy. As of December 2011, India had about 22.4GW installed capacity of renewable technologies-based electricity, exceeding the total installed electricity capacity in Austria by all technologies—according to Wikipedia. However, the penetration of solar PV in India is only 4 per cent of the renewable energy production as of now.

Under the Jawaharlal Nehru National Solar Mission launched by the Indian government, India plans to generate 1 GW of power by 2013 and up to 20 GW grid-based solar power, 2 GW of off-grid solar power and cover 20 million square metres with solar energy collectors by 2020. The country plans utility-scale solar power generation plants through solar parks with dedicated infrastructure by state governments.

There are some roadblocks to overcome in order to realise full solar power potential. Solar plants are expensive as solar cells are still costly. The cost will come down when the efficiency of solar cells can be substan-

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Compensation devices for supplying reactive power to the loads at the load end—rather than reactive power flowing from the generating stations—will reduce the line currents and hence the line losses”

— Prof. (Mrs) Mini Shaji Thomas, Member, IEEE Educational Activities Board, Chair Section outreach committee (SEOC), and Professor & CPIO, Jamia Millia Islamia, New Delhi



tially enhanced.

Land acquisition is another challenge before solar farm developers in India. Some state governments are looking at addressing the problem of land availability through innovation; for example, they are exploring ways to deploy solar capacity above their extensive irrigation canal projects, thereby harvesting solar energy while reducing the loss of irrigation water by solar evaporation.

Thankfully, as volumes in solar plants have picked up, the overall investment in the sector has come down significantly. Three years ago, the amount of power generated through solar power systems was estimated to have cost approximately Rs 15 per unit. Today, this cost has reduced to Rs 7.50—equivalent to 50 per cent cost saving.

Managing pollution along with

power generation is important. An increased focus on energy sources like solar will bring a decrease in pollution, alongside generation of power.

Early last year, India's Bureau of Energy Efficiency (BEE) had launched its 'Perform, Achieve and Trade' (PAT) scheme to set energy efficiency targets for industries. Such schemes will go a long way in helping to improve overall energy efficiency.

Action plan to meet power deficit

Currently, the peak demand shortage in India is around 12 per cent and energy demand shortage is around 11 per cent, as per Central Electricity Authority (CEA) documents.

“The situation demands power deficit to be handled at different levels. As mentioned earlier, managing auxiliary power consumptions plays a vital role in the same. If today installed power capacities amount to 200,000 MW and if we can save at least 2 per cent on auxiliary power consumption, this would give us an additional 4000 MW, which would add up to 1 ultra-mega power project. This is an indication of how we can meet the power deficit situation. It is important that we focus equally on the transmission and distribution of power,” shares Venu.

At the same time, improving efficiency of generation, transmission and distribution alone cannot meet the power deficit. A new generation has to come up to cope with the ever-increasing growth in power and energy demands of this fast-developing country. ●

The author is executive editor at EFY

Electronics makers seek relief from service tax for power sector

Indian Electrical and Electronics Manufacturers' Association (IEEMA), which is the apex representative body of manufacturers of electrical equipment, professional electronics and allied equipment, has extended a request to the government to extend service tax exemption for power generation, transmission and distribution-related services.

In its pre-budget memorandum to the union Finance Ministry, IEEMA reportedly mentioned, “With the mission of providing power to all on demand at an affordable cost, exempting all power projects from the levy of service tax would definitely be a step in the right direction and reduce the project cost.”

According to a Business Standard report, IEEMA also said that it wanted the government to encourage manufacturing and the creation of a level-playing field for the industry in order to maintain competitiveness.

“Higher and accelerated depreciation should be allowed for construction equipment at construction sites at 30 per cent from the present level of 15 per cent so that the equipment can be written off in ten years. This would enable companies to bring in better technology in the manufacturing sector where newer technologies can reduce project cycle time,” IEEMA added in the report. They also requested for refund of excise duty that is paid by domestic manufacturers as a deemed export benefit.

The generation equipment sector is 26 per cent of the total industry, while the transmission and distribution (T&D) equipment sector comprises 74 per cent.